Actuator Force Requirements for Rail Control System

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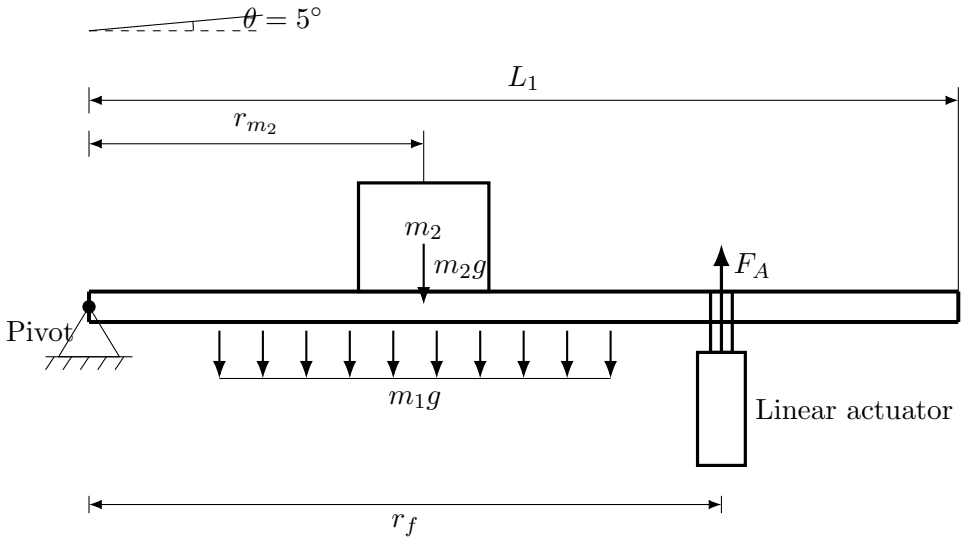
1 System Definition

Define the following parameters:

$$
\begin{aligned}
L_1 &= \text{Length of beam} \\
L_2 &= \text{Length of attached object} \\
m_1 &= \text{Mass of beam} \\
m_2 &= \text{Mass of attached object} \\
r_f &= \text{Actuator attachment distance from pivot} \\
\Delta\theta &= \text{Required angular displacement} \\
t &= \text{Desired motion time} \\
g &= \text{Gravitational acceleration} \\
\Delta z &= \text{Linear distance} \\
\dot{z} &= \text{Linear speed}
\end{aligned}
$$

The beam rotates about the left-hand pivot.

2 Free Body Diagram



where

r_{m_1} = Distance from pivot to beam center of mass
r_{m_2} = Distance from pivot to object center of mass

3 Gravitational Torque

The torque due to gravity is

$$\tau_g = m_1 g r_{m_1} \cos\theta + m_2 g r_{m_2} \cos\theta$$

The maximum gravitational torque occurs at

$$\theta = 0$$

giving

$$\tau_{g,\max} = m_1 g r_{m_1} + m_2 g r_{m_2}$$

4 Angular Acceleration

Assume a symmetric motion profile where the beam starts and ends at rest, accelerating for the first half of the motion and decelerating for the second half.

The angular displacement is

$$\Delta\theta = \frac{\alpha t^2}{4}$$

Thus,

$$\boxed{\alpha = \frac{4\Delta\theta}{t^2}}$$

5 Rotational Inertia

The beam inertia about the rail pivot (uniform rod, pivot at one end) is

$$I_1 = \frac{1}{3} m_1 L_1^2$$

The payload must be taken about the *same* pivot. If the payload centre of mass lies a distance r_{m_2} from the pivot, the parallel-axis contribution is

$$I_2 = I_{2,\mathrm{cm}} + m_2 r_{m_2}^2.$$

For a compact aircraft, $I_{2,\mathrm{cm}}$ is often small compared with $m_2 r_{m_2}^2$, so

$$I_2 \approx m_2 r_{m_2}^2$$

is a useful lower-bound model. Using only $\frac{1}{3} m_2 L_2^2$ about the payload's own end (and omitting $m_2 r_{m_2}^2$) understates inertia when the aircraft sits far out on the rail.

Therefore,

$$\boxed{I_T = I_1 + I_2 = I_1 + I_{2,\mathrm{cm}} + m_2 r_{m_2}^2}$$

6 Acceleration Torque

The torque required to accelerate the rotating system is

$$\tau_a = I_T \alpha$$

Substituting the angular acceleration,

$$\boxed{\tau_a = I_T \left(\frac{4\Delta\theta}{t^2} \right)}$$

7 Total Required Torque

The maximum torque requirement occurs at the beginning of motion where gravity torque is largest and angular acceleration is applied.

$$\tau_T = \tau_{g,\max} + \tau_a$$

Substituting previous expressions,

$$\boxed{\tau_T = m_1 g r_{m_1} + m_2 g r_{m_2} + I_T \left(\frac{4\Delta\theta}{t^2}\right)}$$

8 Actuator Force

If the actuator force acts perpendicular to the beam at a distance r_f from the pivot,

$$\tau_A = F_A r_f$$

Therefore,

$$\boxed{F_A = \frac{\tau_T}{r_f}}$$

Substituting for total torque,

$$\boxed{F_A = \frac{m_1 g r_{m_1} + m_2 g r_{m_2} + I_T \left(\frac{4\Delta\theta}{t^2}\right)}{r_f}}$$

More generally, if the actuator force is not perpendicular to the beam,

$$\tau_A = F_A r_f \sin\phi$$

and

$$\boxed{F_A = \frac{\tau_T}{r_f \sin\phi}}$$

where ϕ is the angle between the actuator force vector and the beam.

9 Linear Distance and Speed

For a vertical actuator mounted beneath the attachment point, the vertical displacement of the attachment point is

$$\boxed{\Delta z = r_f \sin(\Delta\theta)}$$

The corresponding average linear speed is

$$\boxed{\dot{z} = \frac{\Delta z}{t}}$$

It is important to note that Δz represents the vertical motion of the attachment point. The true actuator stroke should be determined from the actuator mounting geometry and may differ from this value if the actuator does not remain vertical throughout the motion.

10 Safety Factor

Including a design safety factor N_s,

$$\boxed{F_{\text{design}} = N_s F_A}$$

where typically

$$N_s = 1.5\text{–}3$$

depending on the application.

11 Ball Screw Motor Requirements

Let

$$\begin{aligned} F_A &= \text{Required actuator force} \\ L_s &= \text{Ball screw lead (linear travel per revolution)} \\ \eta &= \text{Ball screw efficiency} \\ T_m &= \text{Required motor torque} \\ n &= \text{Required motor rotational speed} \end{aligned}$$

The ball screw converts motor rotation into linear motion. The required motor torque is

$$\boxed{T_m = \frac{F_A L_s}{2\pi\eta}}$$

where

- F_A is the actuator force,
- L_s is the ball screw lead,

- η is the mechanical efficiency of the ball screw.

The required motor speed is obtained from the actuator linear speed:

$$\boxed{n = \frac{\dot{z}}{L_s}}$$

where n is expressed in revolutions per second.
Converting to revolutions per minute (RPM),

$$\boxed{n_{\text{RPM}} = 60\left(\frac{\dot{z}}{L_s}\right)}$$

The mechanical power required from the motor is

$$\boxed{P_m = T_m\omega}$$

where

$$\omega = 2\pi n$$

is the motor angular velocity.
Equivalently,

$$\boxed{P_m = \frac{F_A\dot{z}}{\eta}}$$

which provides a useful verification of the motor sizing calculations.

12 Notes on Implementation

- Use a consistent definition of θ (e.g. $\theta = 0$ horizontal) when evaluating τ_g.
- For a triangular velocity profile, peak linear speed is about twice the average $\dot{z} = \Delta z/t$; size motor RPM from the peak unless a softer profile is specified.
- Prototype masses ($\sim 5\,\text{kg}$ stand-in) and full-size Fighter mass ($< 7\,\text{kg}$ MTOW) must not be mixed with outdated placeholders (e.g. 11 kg).
- r_{m_2} is the distance from the *rail pivot* to the payload centre of mass, not $L_2/2$ unless the payload is actually centred that close to the pivot.